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Ensemble of learning transfer models with GAN for data augmentation: Application to skin cancer detection

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Abstract: *Deep Learning models typically require large datasets to achieve high performance, which can be challenging when applied to medical data due to the difficulty in acquiring sufficient observations, particularly when the data is imbalanced between categories like pathological and normal cases. To address this, approaches like transfer learning, ensemble methods, and data augmentation through Generative Adversarial Networks (GANs) can be utilized to improve classification tasks on small datasets. In situations with imbalanced data, resampling techniques are also crucial. This research combines transfer learning with resampling methods to boost the prediction accuracy of minority class samples in small, imbalanced datasets. Additionally, techniques like data retracing and GAN-based augmentation are applied. The dataset used includes small, imbalanced images of skin cancer, aimed at classifying them as malignant or benign. The findings reveal that models using resampling techniques achieve better results, while those without resampling underperform. This underscores the benefit of resampling in enhancing prediction, particularly for the minority class. Furthermore, using GANs for data augmentation improves model performance over those that do not incorporate this technique.*

Keywords: *Convolutional Neural Network; Transfer Learning; Ensemble; Data augmentation.*

Introduction

Convolutional Neural Networks (CNN) are models used to analyze images. But the performance of these models can be affected by the number of observations, for example when there are few observations available for training these models These models require a large amount of data to perform well.

Several studies have been carried out with the aim of solving this problem. For example, Nanni et al. (2022) *et al* present an ensemble with several activation functions for image classification tasks with small data sets. Additionally, Saravanan et al. (2022) uses data augmentation to augment the observations from the database used of brain tumors. Another widely used approach is the use of transfer learning models, for example, Ju et al. (2022) propose a transfer learning model to be used in a small database to identify defects in images.

Another approach that can be used as data augmentation is the generation of synthetic data through Generative Adversarial Network (GAN), according to Chen et al. (2022) synthetic medical images coming from a GAN can be used to supply the small number of images that make up the most diverse medical image banks. Anjos et al. (2020) uses GAN to generate synthetic data and after generation uses the data to diagnose lung cancer.

Recently models involving combinations of transfer learning models have been proposed, Zheng et al. (2022) proposed a model using an Ensemble of Transfer Learning Models (ETCNN) to classify mammography images, the authors also used data augmentation in the training set. Vallabhajosyula et al. (2022) created an automatic detection model for detecting plant diseases, using transfer models and using data augmentation in the training set.

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